Modelling of a PMUT Pulse-Echo Through an Oil-Filled Steel Tube

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Abstract

Piezoelectric micromachined ultrasonic transducers (PMUTs) are usually designed for short-range imaging in soft tissue near 1 MHz. I ask a different question: can a PMUT range a target by pulse-echo down a 3 m oil-filled steel tube, and what actually limits it? Starting from the published system-level transducer model of Dangi and Pratap (2017), which radiates into air over sub-half-metre distances and carries no noise model, I add the missing pieces: an acoustic-channel model for the tube treated as a waveguide (transfer-matrix wall coupling, confinement instead of spherical spreading, frequency dependent oil absorption, and a near-total far-end reflector), a first-principles charge-amplifier receiver-noise model, and a two-dimensional finite-difference wave simulation for visualization. Three results emerge that overturn the default intuition. First, the conventional choice of frequency is backwards: oil absorption scales as f^2 over the fixed round trip, so the echo amplitude collapses as $\exp(-kf^2)$, giving 1920 dB of round-trip loss at 1 MHz versus 120 dB at 250 kHz; the usable band lies near and below 135 kHz, not at 1 MHz. Second, when transmit and receive are forced to derive from one electromechanical coupling (so the two-way link is passive, which the published sensitivities are not: they imply voltage gain above unity), the detection boundary sits at roughly 135 kHz for a representative 20 dB/m oil. Third, the binding constraint is the receiver, not the channel: a thin-film-PZT clamped capacitance of order 479 nF feeding a low-noise charge amplifier sets a noise floor tens of dB above the convenient "10 μV" placeholder. I give the link budget, the reciprocity relation, the noise model, a 100/135/200 kHz comparison spanning detectable to buried, and matching wave-propagation movies. Absolute signal-to-noise carries order-of-magnitude uncertainty because the source paper's own transmit and receive sensitivities disagree by more than $100\times$; the relative structure, the cliff, and the receiver-limited regime are robust.

1 Introduction

Ultrasonic ranging through metal walls and along fluid-filled tubes is attractive for level sensing, leak and blockage detection, and through-wall telemetry in sealed vessels. Piezoelectric micromachined ultrasonic transducers (PMUTs) are small, low-power, and CMOS-compatible, which makes them appealing for such embedded sensing [Jung et al., 2017, Przybyla et al., 2011]. Almost all PMUT design practice, however, targets short-range medical and fingerprint imaging, where the operating frequency is pushed toward 1 MHz and above to improve axial resolution, and where the propagation medium is soft tissue or water over millimetres to a few centimetres.

The question that motivates this work is whether that design point transfers to a very different regime: a transducer at one end of a 3 m oil-filled steel tube, attempting to detect the echo from the far end. The starting point is the system-level analytical model of Dangi and Pratap [2017], which compactly relates a multilayer PMUT's transmit pressure, receive charge, and quality factor to its geometry through three nearly invariant mode-shape factors. That model is validated for radiation into air over distances below 0.5 m and contains no treatment of layered solid/liquid channels, of propagation to metres, of pulse-echo, or of noise. It therefore cannot, by itself, answer the ranging question.

I extend it into an end-to-end link model and report three findings. (1) The operating frequency is set by a steep absorption cliff: because the oil attenuation coefficient grows as f^2, the round-trip loss over 6 m is 11 dB at 75 kHz, 120 dB at 250 kHz, and 1920 dB at 1 MHz; the intuitive high-frequency choice is not merely suboptimal but fatal, and the usable band is far lower. (2) Confinement, not radiation, governs feasibility: the tube is a waveguide, so the spherical spreading that dominates a free-field budget is replaced by a bounded cross-section fill loss. (3) When the transmit and receive paths are required to come from a single coupling so the link is passive, and noise is modeled from the transducer capacitance and amplifier rather than assumed, the detection boundary lands near 135 kHz and is set by the receiver electronics rather than by the acoustic channel.

My contribution is a careful, reciprocity-consistent system model and the resulting design inferences, not a new transducer or a new experiment. I am explicit about a calibration caveat: the source paper's published transmit and receive sensitivities are mutually inconsistent by more than two orders of magnitude, so absolute signal-to-noise ratios (SNRs) are uncertain to within roughly that factor. The robust outputs are the relative trends, the f^2 cliff, the passivity constraint, and the receiver-limited regime, all of which are independent of the absolute calibration.

2 Background: the starting transducer model

A PMUT is a clamped circular multilayer plate with a piezoelectric layer (here lead zirconate titanate, PZT) between electrodes on a passive structural layer. Applying a voltage produces an unbalanced in-plane stress and hence a bending moment, so the plate vibrates flexurally and pumps the adjacent fluid; conversely an incident pressure flexes the plate and generates charge. Dangi and Pratap [2017], building on the natural-frequency derivation of Sammoura et al. [2012], reduce this to a lumped electro-mechano-acoustic circuit and obtain closed forms for the first-mode natural frequency, the receive charge sensitivity (their Eq. 17), the transmit response, and the quality factor, all written through three mode-shape participation factors $\Lambda', \Lambda_1, \Lambda_2$ that vary little across designs. I reuse those relations for the transducer and treat everything outside the device, the channel and the receiver electronics, as the part to be built. Section 3.2 returns to a subtlety in how the transmit and receive sides of that model must be coupled.

3 System model

Figure 1 shows the geometry. A PMUT (or, more realistically, a small PMUT array forming an effective aperture of radius a_{ap}) sits at the near end of a steel tube of outer diameter 300 mm and wall thickness 10 mm, so the inner oil radius is $R_{\mathrm{in}} = 140$ mm. The tube is filled with a synthetic mineral oil and is $L = 3$ m long. The far end is a thick steel block, thick enough that the steel/air interface behind it is irrelevant, so it acts as a near-total acoustic reflector. The transducer both transmits and, after the round trip, receives.

3.1 Acoustic channel as a link budget

I track acoustic intensity through the round trip as a product of factors, then convert to an incident pressure at the receiver. The transmitted power crosses the near steel cap into the oil, propagates and attenuates to the far end, reflects, returns, and crosses the cap again. Writing each stage as an intensity factor,

$$I_{\mathrm{back}} = I_{\mathrm{face}} \cdot \underbrace{\tau_{\mathrm{wall}}}_{\text{out}} \cdot \underbrace{\left(\frac{a_{\mathrm{ap}}}{R_{\mathrm{in}}}\right)^2}_{\text{fill}} \cdot \underbrace{\Gamma^2}_{\text{far end}} \cdot \underbrace{e^{-2\alpha(f)\,(2L)}}_{\text{oil absorption}} \cdot \underbrace{\tau_{\mathrm{wall}}}_{\text{return}} . \tag{1}$$

Three of these terms carry the physics. The wall term τ_{wall} is the intensity transmission of the 10 mm steel layer, computed from the standard single-layer acoustic transfer matrix between the transducer-side medium and the oil; for a well-coupled (bonded) transducer it is a flat -10 dB per crossing, while for a dry, air-backed transducer it develops deep resonance windows at multiples of $c_{\mathrm{steel}}/2d$. The confinement term replaces free-field spherical spreading: in an unbounded fluid the round-trip on-axis pressure falls as the square of $a^2/(\lambda z)$, costing about -41 dB at 3 m, but in the

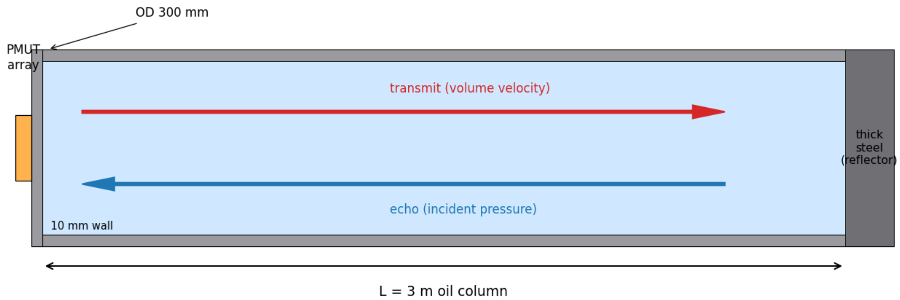

Figure 1: Pulse-echo geometry. A PMUT array at the near end launches into the oil column through the 10 mm steel end-cap; the wave travels 3 m, reflects off the thick-steel far end, and returns to the same transducer. The tube walls confine the beam, so the channel is a waveguide rather than free space.

tube the energy that fills the cross-section is conserved and the only geometric loss is the fraction $(a_{\rm ap}/R_{\rm in})^2$ that the receive aperture re-samples, here -28.9 dB, and falling to 0 dB as the aperture fills the tube. The absorption term uses the classical thermoviscous law [Kinsler et al., 2000] in which the coefficient grows quadratically with frequency,

$$\alpha_{\rm dB}(f) = \alpha_0\,(f/f_{\rm ref})^2, \qquad \alpha_0 = 20\text{ dB/m at } f_{\rm ref} = 250\,\text{kHz}. \tag{2}$$

This single term dominates the frequency dependence of the whole system and is the source of the result in Section 5.1. A small additional waveguide wall-bounce loss of 0.5 dB/m is included. The incident pressure at the receiver is $P_{\rm inc} = \sqrt{2Z_{\rm steel}I_{\rm back}}$.

3.2 Reciprocity-consistent transmit and receive

The error that is easy to make, and that I initially made, is to anchor the transmit and receive paths independently: transmit from an assumed launched power, receive from the published charge sensitivity. Doing so lets the end-to-end voltage transfer exceed unity at low frequency (in my first runs, a 140 V echo from a 10 V drive), which is impossible for a passive two-way link. The fix is to require both directions to come from one electromechanical coupling. In the lumped model the transmit volume-velocity response and the receive charge sensitivity share the transformer ratio ϕ and the mechanical impedance Z_m, which forces the reciprocity relation

$$\left.\frac{\dot q}{V_{\rm in}}\right|_{\rm transmit} = \omega_{00}\left.\frac{Q}{P_{\rm in}}\right|_{\rm receive}, \tag{3}$$

where $\dot q$ is the volume velocity, ω_{00} the resonance, and $Q/P_{\rm in}$ the charge per unit incident pressure. I anchor the absolute scale to the paper's *measured* transmit (a 1 mm, 126 kHz device deflects about 3 nm/V at resonance, their Fig. 9) and *derive* the receive sensitivity from Eq. (3), rather than using their Eq. 17 in isolation. The aperture is treated as an array of elements each sized to resonate at the operating frequency; the element count cancels in the SNR, so the result is aperture self-consistent. With this construction the implied transmit is physically sane (55 mW launched, face velocity 5.5 mm/s, face pressure 254 kPa), and the two-way transfer never exceeds $V_{\rm echo}/V_{\rm in} = 0.77$ across the band, restoring passivity. The reciprocity-derived receive sensitivity at 250 kHz is 27 fC/Pa, about $41\times$ smaller than the paper's Eq. 17 value of 1130 fC/Pa; the two published numbers are not mutually consistent, which is the origin of the calibration caveat stated throughout.

3.3 Receiver noise

The received charge is read by a charge amplifier with feedback capacitance C_f. The dominant noise is the amplifier input voltage noise e_n multiplied by the capacitive noise gain $(C_{\rm in}/C_f)$, where $C_{\rm in} = C_s + C_{\rm stray} + C_f$ and C_s is the PMUT clamped capacitance. I add the amplifier current noise through the feedback transimpedance and the PZT dielectric-loss (Johnson) noise, and integrate over the resonant equivalent noise bandwidth $\text{ENBW} = (\pi/2)(f_0/Q)$:

$$V_{\rm noise} = \sqrt{(e_n\,C_{\rm in}/C_f)^2 + (e_i\,Z_f)^2 + 4k_BT\,G\,Z_f^2}\ \sqrt{\text{ENBW}}. \tag{4}$$

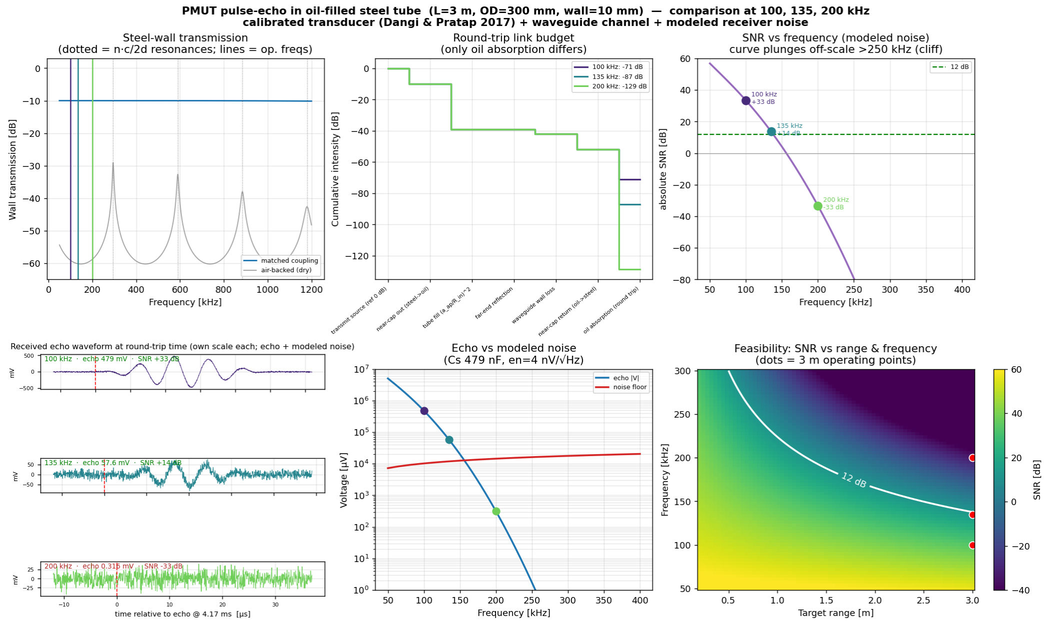

Figure 2: Calibrated comparison at 100/135/200 kHz. (1) Steel-wall transmission: flat for matched coupling, resonant if dry. (2) Round-trip link budget; only the oil-absorption term differs between frequencies. (3) Absolute SNR versus frequency with the three operating points marked; the curve plunges off-scale above 250 kHz. (4) The actual received echo waveform plus modeled noise at each frequency, each on its own absolute scale: a clean burst at 100 kHz, an echo emerging from noise at 135 kHz, and pure noise at 200 kHz. (5) Echo voltage and the modeled noise floor versus frequency, crossing near 135 kHz. (6) Feasibility map of SNR over range and frequency, with the 12 dB contour and the 3 m operating points.

Thin-film PZT (650 nm thick) over a 10 mm aperture gives a large clamped capacitance of order 479 nF, so with a low-noise $e_n = 4\, nV/\sqrt{\mathrm{Hz}}$ the voltage-noise term dominates and the noise floor is of order tens of millivolts referred to the amplifier output, far above a convenient 10 μV placeholder. The link is therefore receiver-limited.

4 FDTD visualization

To see the propagation I built a two-dimensional finite-difference time-domain solver on an axial slice of the tube, following the explicit grid approach of WaveSimulator2D [0x23 (Michael Schardt), 2022] but using a variable-density velocity-pressure staggered scheme [Virieux, 1986] so that reflections follow the true acoustic impedance $Z = \rho c$. The oil sound speed (1440 m/s) is exact, so the round-trip time-of-flight of about 4.17 ms is physical. To keep the time step affordable, the steel sound speed is reduced while its density is kept high, which preserves the impedance contrast and hence the strong reflections; this is a visualization trade-off and the SNR numbers come from the link budget of Section 3, not from the FDTD. Each rendered movie has three synchronized panels: the wave field, the transmitter drive, and the received signal with the modeled noise, plus a running readout of physical simulation time.

5 Results

Figure 2 compares the calibrated model at 100, 135, and 200 kHz, the frequencies that bracket the detection boundary.

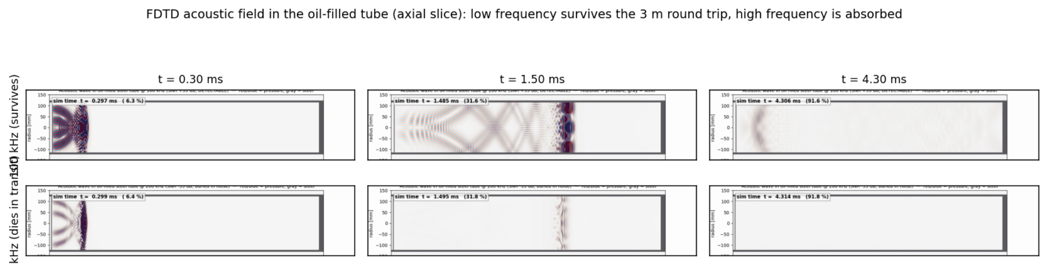

Figure 3: FDTD acoustic field on an axial slice. Top: at 100 kHz the pulse travels the full 3 m, reflects, and returns (echo present near the near end at 4.30 ms). Bottom: at 200 kHz the pulse is absorbed before completing the round trip, so no echo returns. The contrast is the f^2 absorption cliff made visible.

5.1 The f^2 absorption cliff sets the frequency

Because the absorption coefficient in Eq. (2) grows as f^2, the dB loss over the fixed 6 m round trip also grows as f^2 and the echo amplitude collapses as $\exp(-kf^2)$. The round-trip oil loss is 11 dB at 75 kHz, 120 dB at 250 kHz, and 1920 dB at 1 MHz. The receive sensitivity grows only as $1/f^2$ and the noise as $\sqrt{f}$, so neither offsets the collapse. The practical consequence is stark and runs against ultrasonic-imaging intuition: a frequency chosen for axial resolution is exactly the wrong choice for long-range ranging in an absorbing liquid, and the usable band sits near and below 135 kHz. Panel 3 of Figure 2 shows the cliff directly; the curve is bounded to a readable range because past 250 kHz it descends through hundreds of negative dB.

5.2 A detection boundary near 135 kHz

With the reciprocity-consistent transducer and the modeled noise floor, the absolute SNRs at the three operating points are +14 dB at 135 kHz, +33 dB at 100 kHz, and -33 dB at 200 kHz, against a 12 dB detection threshold. The received-waveform panel (Figure 2, panel 4) tells the story in the time domain on per-frequency absolute scales: the 100 kHz echo is a clean 479 mV burst well above noise, the 135 kHz echo (57.6 mV) just clears the noise (the knife-edge case), and the 200 kHz echo (0.316 mV) is buried. Figure 3 shows the corresponding FDTD fields: at 100 kHz a wave packet survives the round trip and returns to the near end, while at 200 kHz it is absorbed in transit and never returns. The two views, a link budget and an independent wave solve, agree on the boundary.

5.3 The receiver, not the channel, is the limit

The modeled noise floor is dominated by the amplifier voltage noise times the large thin-film-PZT capacitance (Eq. (4)), giving an output-referred floor of order tens of millivolts. This is what determines detectability: the same channel against a naive $10\,\mu$V floor would appear roughly 70 dB more favorable. The levers that move the boundary are therefore on the receiver and the frequency, namely a lower transducer capacitance (thicker or bulk PZT, or charge matching at the element), a quieter front end, a narrower bandwidth or coded excitation, and above all a lower operating frequency to escape the absorption cliff. A range-by-frequency feasibility map (Figure 2, panel 6) shows the 12 dB contour crossing roughly 135 kHz at 3 m and opening up as either the range shortens or the frequency drops.

5.4 Why confinement matters

A free-field version of the same budget, target at 3 m in open oil, is dominated by spherical spreading and is at best marginal and strongly dependent on whether the target is a large specular reflector or a small scatterer. The tube removes that term: the wave is guided, the far end reflects nearly all of it, and the geometric penalty reduces to the aperture fill fraction. This is why a sealed oil-filled tube is a far more forgiving channel than open water, and why the feasibility question turns on absorption and electronics rather than on spreading.

6 Limitations and failure modes

The dominant limitation is absolute calibration. The source paper's transmit deflection sensitivity, its Eq. 17 receive sensitivity, and its tabulated design-map sensitivities disagree by more than $100\times$ when cross-checked, so any absolute SNR inherits roughly that uncertainty. I therefore present the absolute numbers as indicative and rest the conclusions on the relative structure, which is calibration independent: the f^2 cliff, the passivity constraint, the capacitance-limited noise regime, and the existence and approximate location of a detection boundary. The FDTD is a two-dimensional axial slice with a reduced steel speed and is used only for visualization. The oil attenuation and PZT permittivity are representative values, not measurements of a specific oil or device. There is no physical experiment; this is a modeling study whose purpose is to set design intuition and to provide a self-consistent, reproducible budget that a measurement campaign can later anchor.

7 Conclusion

For PMUT pulse-echo down a 3 m oil-filled steel tube, the conventional imaging design point near 1 MHz fails by a wide margin, because oil absorption over the round trip grows as f^2 and annihilates the echo. Confinement in the tube, not free-field radiation, makes the channel favorable, and once transmit and receive are tied to one coupling so the link is passive, the binding constraint is the receiver capacitance and amplifier noise rather than the acoustics. The usable band is near and below 135 kHz, and the design effort should go into lowering the operating frequency and the receiver noise floor. The model, the reciprocity relation, the noise budget, and the visualizations are reproducible from the released code.

8 Code and data availability

All code, figures, and the wave-propagation videos are released at `https://github.com/rayjyotir5/pmut-tube-echo`. The repository contains the 1D link-budget and noise model (`pmut_echo_sim.py`), the 2D FDTD solver and animation generator (`pmut_wave2d.py`), the figure scripts, the rendered 100/135/200 kHz movies, and a README with step-by-step reproduction instructions. The results figure and reports regenerate with `python3 pmut_echo_sim.py 100 135 200`; each FDTD movie with `python3 pmut_wave2d.py <freq_kHz>`.

Acknowledgements

This study, including the derivations, the simulation code, the figures, and the writing of this paper, was carried out with Claude (Anthropic) acting as an autonomous research and writing assistant under human direction. The human contribution was the choice of the starting paper, the physical scenario and a small set of physical parameters, and steering and quality feedback; the AI contribution was the reading, modeling, coding, debugging, figure generation, and drafting. A full provenance record appears in Appendix A.

References

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A Provenance: human input versus AI effort

This work was produced in a single interactive session in which a human supplied the starting paper, the physical scenario, a small set of physical parameters, and steering and quality feedback, while an AI agent (Claude, Anthropic, Opus 4.8) performed all reading, derivation, coding, debugging, figure generation, and writing. I log this explicitly so the division of labor is preserved.

A.1 Verbatim human prompts

Light typographic errors are preserved as written.

1. "read the modelling.pdf and understand what it is doing, will specify the actual problem after that"
2. "Has analytical model shown acoustic transmission through 10 mm steel + synthetic mineral oil yields detectable echo at 3 m with SNR $\geq$ 12 dB at 1 MHz centre frequency? if not i want to set up a simulation and visualize the results" (clarifying choices: pulse-echo through wall; sweep noise; 1D analytical + transfer matrix)
3. "lets now upgrade the sim to a 2d environment, a steel tube is 3m long, 10mm thick and since it is the cross section of a circular tube, the outer dia of the tube is 300mm. the pmut is positioned at one end and the other end forms the reflecting surface (this end may have increased thickness of steel so as to ignore steel-air interface). tube is filled with oil. now run the simulation and repeat previous sweeps. feel free to re-center the frequency sweep and not assume 1MHz is a fixed input so that SNR is optimal. i'd also like to see a video or gif animation, use the WaveSimulator2D approach if required."
4. "oil attenuation is 20 db/m and waveguide wall-bounce loss of 0.5db/m is fine. in the mp4, please render two more panels below the wave propagation animation. one is the pmut transmitter signal over time and a vertical line moves along synced to wave animation. similarly a panel for measured pmut receiver signal. also have a legend to show physical time of sim progress. hope the wave is simulated with noise considered at 250khz"
5. "erm 20db/m was not at 1mhz, it is at 250khz"
6. "render a second clip at the ∼50-100 kHz optimum so you can see a clean, surviving echo for contrast"
7. "is the sharp drop at 1MHz in the SNR vs frequency correct? why is this happening? would it help to now stop the sweep and model the noise correctly?"
8. "yes" (proceed to tie transmit and receive to one reciprocity-consistent coupling)
9. "lets now regenerate the results plots and wave propagation animations with the new calibrated model and view it above and below 135kHz"
10. "make sure to regenerate pmut_tube_results.png"
11. "i dont want three png files, i want a single one"
12. "the plots are at 250khz, pls update to 100, 135, 200"
13. "plot y axes are not scaled correctly, getting cropped"
14. "panel 1 and 4 as well, will log scale help?"
15. "i dont like the bar chart in panel 4, id rather see the waveform, not normalized to noise"
16. "treat this entire session as the research done, document everything and produce the resultant paper"

A.2 Session flow

Turn 1 established what the source model does. Turn 2 established that it does not address the ranging question and produced a first free-field link budget (two self-found bugs fixed: double-counted spreading, and an unphysical multi-watt transmit). Turn 3 reformulated the channel as a tube waveguide and added the 2D FDTD (a sign-error instability was found and fixed). Turns 4 to 6 set parameters, built the synchronized three-panel animation, corrected the attenuation reference frequency (a 112 dB change at the operating point), and rendered a low-frequency contrast clip. Turn 7 diagnosed the SNR cliff as the f^2 absorption law and replaced the assumed noise floor with a modeled one. Turn 8 tied transmit and receive together by reciprocity, removing an impossible gain-above-unity artifact. Turns 9 to 15 regenerated all outputs with the calibrated model and refined figure quality. Turn 16 produced this paper.

B Full parameter set

All values are documented in the released configuration (`src/pmut_echo_sim.py`).

Group	Quantity	Value
Geometry	tube length L	3.0 m
	outer diameter / wall	300 mm / 10 mm
	inner oil radius R_{in}	140 mm
	effective aperture radius a_{ap}	5 mm
Steel	density / speed	7850 kg/m^3 / 5900 m/s
	impedance	46.3 MRayl
Oil	density / speed	850 kg/m^3 / 1440 m/s
	impedance	1.22 MRayl
	absorption α_0 at 250 kHz	20 dB/m, $\propto f^2$
	waveguide wall-bounce loss	0.5 dB/m
	far-end reflection Γ	0.99
Transducer	quality factor Q	55
	$\Lambda', \Lambda_1, \Lambda_2$	0.99, 0.33, 0.20
	$e_{31,f}$	-24.24 C/m^2
	flexural rigidity D_e (calibrated)	1.0×10^{-5} N m
	plate eigenvalue α_{00}	3.19
	deflection sens. anchor	3 nm/V at 500 μm, 126 kHz
Receiver	feedback cap C_f	10 pF
	amplifier voltage noise e_n	4 nV/$\sqrt{\mathrm{Hz}}$
	PZT permittivity / thickness	1000 / 650 nm
	electrode fill / stray cap	0.7 / 5 pF
	dielectric loss $\tan\delta$	0.02
Operating	drive voltage V_{in}	10 V
	detection threshold	12 dB

Table 1: Model parameters. Steel speed is reduced to 2500 m/s in the FDTD only (density kept), to relax the time step while preserving the impedance contrast.

C Derivations

Wall transfer matrix. The intensity transmission of a single steel layer of impedance Z and thickness d between a source medium Z_1 and the oil Z_2 is

$$\tau_{\mathrm{wall}} = \frac{4Z_1Z_2}{(Z_1+Z_2)^2\cos^2(kd) + \left(Z + Z_1Z_2/Z\right)^2\sin^2(kd)}. \tag{5}$$

For a transducer well coupled to steel ($Z_1 = Z$) this reduces to the bare interface value $4Z_1Z_2/(Z_1+Z_2)^2 \approx -10$ dB, independent of frequency; for an air-backed transducer ($Z_1 = Z_{\mathrm{air}}$) the $\sin^2$ term creates deep nulls between the half-wave resonances at $n\,c_{\mathrm{steel}}/2d$.

Confinement (fill) factor. In free space the on-axis far-field pressure of a piston of radius a falls as $\pi a^2/(\lambda z)$, so the monostatic round trip costs that factor squared, about -41 dB at $z = 3$ m. In the tube the launched power instead fills the cross-section and is conserved (apart from absorption); the only geometric loss is the fraction the receive aperture re-samples on return, $(a_{\text{ap}}/R_{\text{in}})^2$, which tends to unity as the aperture fills the tube.

Reciprocity relation. In the lumped circuit the transformer (ratio ϕ) maps the electrical port to the mechano-acoustic port with $P = \phi V$ and $I = \phi \dot{q}$. Transmit gives $\dot{q} = \phi V_{\text{in}}/Z_m$; receive gives $Q = \phi P_{\text{in}}/(j\omega Z_m)$. Eliminating ϕ and Z_m yields Eq. (3). Anchoring $\dot{q}/V$ to the measured 3 nm/V deflection and deriving Q/P_{in} from it makes the two paths share one coupling, so the two-way transfer cannot exceed what passivity allows.

Noise. The output-referred voltage-noise gain of a charge amplifier reading a source capacitance C_s is C_{in}/C_f with $C_{\text{in}} = C_s + C_{\text{stray}} + C_f$. The three terms in Eq. (4) are amplifier voltage noise, amplifier current noise through the feedback transimpedance $Z_f = 1/(\omega C_f)$, and the PZT dielectric-loss Johnson noise with parallel conductance $G = \omega C_s \tan\delta$, all integrated over $\text{ENBW} = (\pi/2)(f_0/Q)$.

FDTD update. The variable-density acoustic field uses a staggered velocity-pressure leapfrog: $\mathbf{v} -= (\Delta t/\rho)\nabla p$, then $p -= \Delta t\, \rho c^2\, \nabla\cdot \mathbf{v}$, stable for $c_{\max}\Delta t/\Delta x \leq 1/\sqrt{2}$ in 2D. Open boundaries use a graded sponge layer; oil absorption is applied as a per-step decay $\exp(-\alpha c\, \Delta t)$ matched to Eq. (2) at the operating frequency. A sign error in the velocity update (using $+\nabla p$) made the scheme blow up; correcting it to $-\nabla p$ restored stability.

D Additional results

f (kHz)	oil loss (dB, RT)	echo	noise floor	SNR (dB)	$V_{\text{echo}}/V_{\text{in}}$
75	11	–	–	+46	0.168
100	19	479 mV	10.2 mV	+33.4	0.048
135	35	57.6 mV	11.9 mV	+13.7	0.0058
200	77	0.316 mV	14.5 mV	−33.2	3.2×10^{-5}
250	120	1.7 μV	16.2 mV	−79	1.8×10^{-7}
1000	1920	$\sim$0	32 mV	−1898	4×10^{-98}

Table 2: Calibrated link at the design aperture (5 mm) and 3 m range. The detection boundary (12 dB) sits near 135 kHz. $V_{\text{echo}}/V_{\text{in}} < 1$ at every frequency confirms the model is passive after the reciprocity fix; before the fix it reached 14 at 75 kHz.

E Calibration inconsistency

The source paper reports three sensitivity figures that do not agree when cross-checked. Its Eq. 17 receive charge sensitivity evaluates to about 1130 fC/Pa for the configured stack, while the value implied by its measured 3 nm/V transmit deflection through the reciprocity relation is about 27 fC/Pa, a factor of 41. Separately, its design-map deflection sensitivity (Table 3, $\sim$0.42 μm/V) and its measured deflection (Fig. 9, 3 nm/V) differ by more than $100\times$. I therefore treat absolute SNR as uncertain to roughly two orders of magnitude and rest the conclusions on the calibration-independent structure: the f^2 cliff, passivity, the capacitance-limited noise regime, and the existence of a detection boundary. Under any self-consistent choice within this spread, 250 kHz and above fail and the usable band is at low frequency.

F Reproducibility

```
cd src
python3 pmut_echo_sim.py 100 135 200    # -> ../figures/pmut_tube_results.png + reports
python3 pmut_wave2d.py 100              # -> 100khz.mp4/.gif  (also 135, 200)
```

```
python3 make_paper_figures.py          # -> schematic + wave montage
cd ../paper
pdflatex main && bibtex main && pdflatex main && pdflatex main
```

Environment: Python 3.11, NumPy 2.0, SciPy 1.14, Matplotlib 3.10, imageio with the ffmpeg plugin; TeX Live 2025. The 1D link budget runs in seconds; each FDTD clip is a few minutes on a single CPU core.

G AI involvement checklist (summary)

Research stage assessment: ideation and framing were human-seeded (the scenario and the ranging question) and AI-developed; the literature anchor (one paper) was human-provided; experiment design, implementation, analysis, and writing were AI-led with human steering and feedback (High AI involvement). AI systems used: Claude (Anthropic), Opus 4.8, as an agent with code execution and figure generation. Known limitations and failure modes are stated in the main text (Limitations) and Appendix E: absolute calibration uncertainty, a visualization-only 2D solver, representative rather than measured material properties, and the absence of a physical experiment.